

TASCA 3

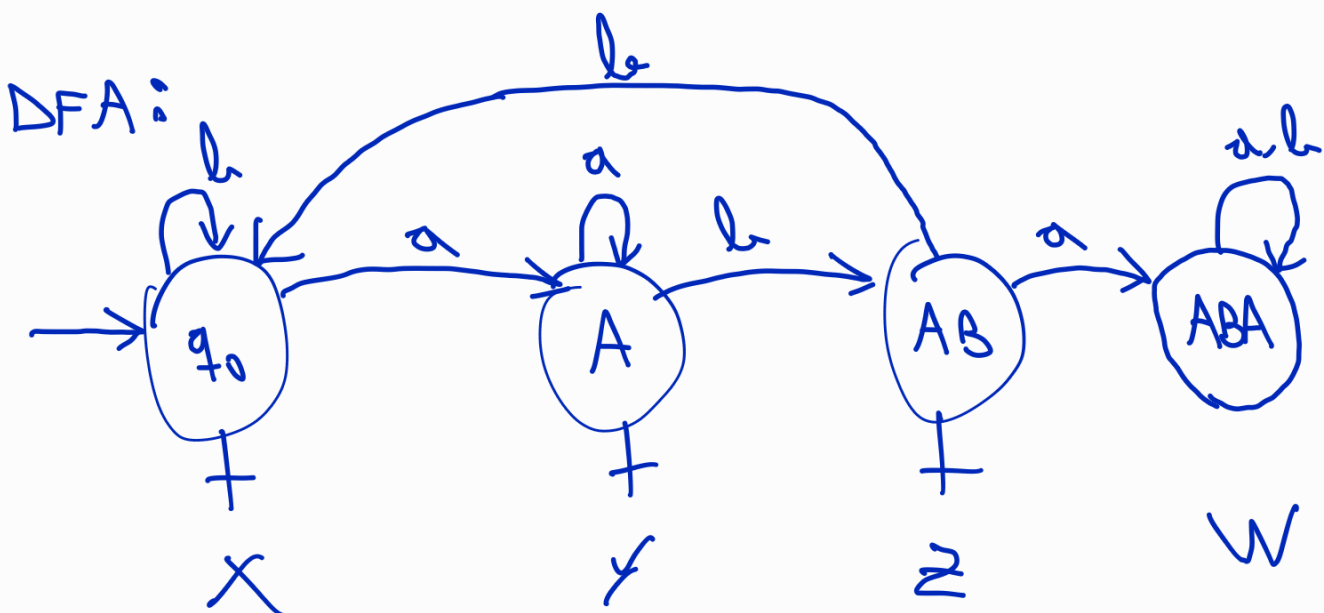
3.9.3.d) Per cada un dels llenguatges següents, doneu

un DFA A_L que reconegui L i dues expressions regulars que representin L . Obtinguen les expressions regulars fent servir el lema d'Arden i la versió simètrica del lema d'Arden sobre A_L on

d) L és el llenguatge dels mots sobre $\{a, b\}$ que no contenen el submot aba :

• Lema Arden: $X = AX + B \Rightarrow X = A^*B$ si $1 \notin A$

• Lema Arden Simètric: $X = XA + B \Rightarrow X = BA^*$



: Derivatem

$$L_{q_0} = x$$

$$L_A = y$$

$$L_{AB} = z$$

$$L_{ABA} = w$$

Expremier Regular 1 : Lemma d'Arden (Dreta)
(a or anan)

$$\left\{ \begin{array}{l} x = \lambda + aY + bX \\ y = \lambda + aY + bZ \\ z = \lambda + aW + bX \\ w = aW + bW \end{array} \right. \quad \begin{array}{l} w = (a+b)W + \emptyset \\ w = (a+b)^* \cdot \emptyset = \emptyset \end{array}$$
$$\left[L(\text{DFA}) = X + Y + Z \right]$$

$$y = \lambda + aY + bZ$$
$$\Downarrow$$

$$y = aY + (\lambda + bZ)$$
$$\Downarrow \text{Arden}$$

$$y = a^*(\lambda + bZ)$$
$$\Downarrow \text{Subst } aX$$

$$x = \lambda + aa^*(\lambda + bZ) + bX$$

$$x = bX + (\lambda + aa^*(\lambda + bZ))$$
$$\Downarrow \text{Arden}$$

$$x = b^*(\lambda + aa^*(\lambda + bZ))$$
$$\Downarrow \text{Subst } aZ$$

$$z = \lambda + aW + bb^*(\lambda + aa^*(\lambda + bZ))$$

$$z = \lambda + aW + bb^*(\lambda + aa^* + aa^*bZ)$$

//

$$Z = 1 + aW + bb^* + bb^*aa^* + bb^*aa^*bZ$$

$$Z = bb^*aa^*bZ + (1 + aW + bb^* + bb^*aa^*)$$

$\Downarrow$ Arden

$$Z = (bb^*aa^*b)^* (1 + aW + bb^* + bb^*aa^*)$$

$\Downarrow W = \emptyset$

$$Z = (bb^*aa^*b)^* (1 + bb^* + bb^*aa^*)$$

$$X = b^* (1 + aa^* (1 + b(bb^*aa^*b)^* (1 + bb^* + bb^*aa^*)))$$

$$Y = a^* (1 + b(bb^*aa^*b)^* (1 + bb^* + bb^*aa^*))$$

* loops

+ caminos

Dreta

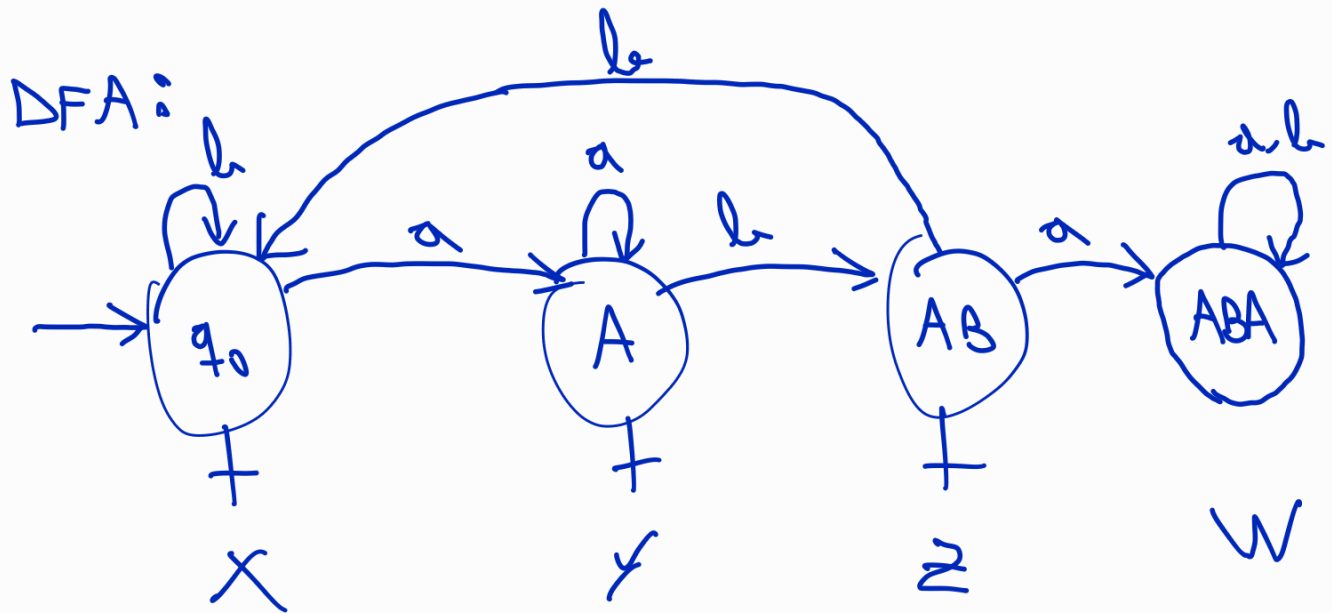
aB

(a on veiny)

Esquerra

Ba

(Com aniden o mi)



Per simètrie d'Arden $X = XA + B \Rightarrow X = BA^*$

Esquerra
can arribem a X

$$\begin{cases} X = 1 + Xb + Zb \\ Y = Xa + Ya \\ Z = Yb \\ W = Za + Wa + Wb \end{cases}$$

$$L(DFA) = X$$

$$Y = Ya + Xa$$

$\Downarrow$ Arden

$$Y = Xaa^*$$

$\Downarrow$ Substitueix Z

$$Z = Xaa^*b$$

$\Downarrow$ Substitueix X

$$X = 1 + Xb + Xaa^*bb$$

Arden

$$X = Xb + (1 + Xaa^*bb)$$

$$X = (1 + Xaa^*bb)b^*$$

$$\Downarrow$$

$$X = b^* + Xaa^*bb^*b^*$$

$$X = Xaa^*bb^*b^* + b^*$$

$\Downarrow$ Arden

$$\boxed{X = b^*(aa^*bb^*b^*)^*}$$

$$3.1.2 c) L = \{a^m b^n \mid m, n \in \mathbb{N} \wedge m \neq n\}$$

Demuestre que no es regular:

Sup L regular $M(Q, \Sigma, \delta, q_0, F) \mid L(M) = L$

$|Q| = p \Rightarrow$ todas las palabras $w \in L$ con $|w| \geq p$
 se "repiten estados"

$$\forall p \geq 0 \quad \exists y^i z \quad S = a^{p+1} b^p \quad n=p \quad n=p+1$$

$$|xy| \leq p, |y| \geq 1, \forall i, i \geq 0 \quad xy^i z \in L$$

$$y = a^k \quad S = a^{(p+1) + (i-1)k} b^p$$

$$k \geq 1$$

$$i \geq 0$$

$$S = a^1 b^1$$

Contradicción!

Converte

$$w = a^p b^N$$

$$w = xyz \begin{cases} |xy| = p \\ |y| \geq 1 \end{cases}$$

$$xy^iz \notin L$$

||

$$a^l a^k a^{p-l-k} b^N$$

$$a^{p+(i-1)k} b^N \Rightarrow N = p + (i-1)k$$

$$\left[i = \frac{N-p}{k} + 1 \right]$$

$$\downarrow \text{natural } i \leq k \leq p$$

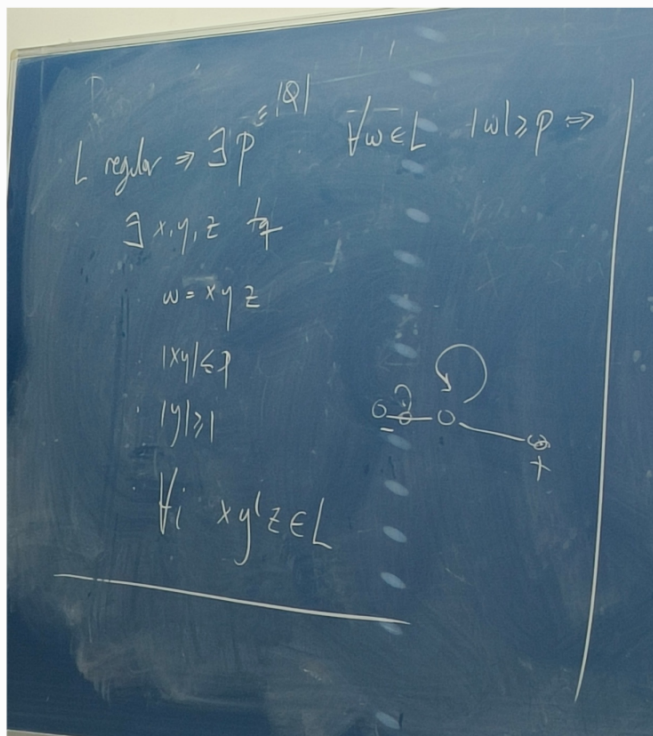
$$N = p! + p$$

($i \leq k \leq p$) (p natural)

Alter

$$\bar{L} \cap \{a^m b^n\}^* = \{a^m b^n \mid m \geq 0\}$$

Bonbes



$$x = a^l$$

$$y = a^k$$

$$z = a^{p-l-k} b^N$$

3.1.3 a) Demonstra que $L = \{c^m a^n b^m \mid m, n \in \mathbb{N}\} \notin A_{reg}$

Per reducció a l'absurd: suposem que $L \in Reg$

Definim h
$$\begin{cases} h(a) = a \\ h(b) = b \\ h(c) = \epsilon \end{cases} \quad L \notin Reg \rightarrow h(L) \notin Reg$$

ja e veiem que
no e regular
 $\uparrow$

$$h(L) = \{h(c^m a^n b^m) \mid m, n \in \mathbb{N}\} = \{a^n b^m \mid m, n \in \mathbb{N}\}$$

$\notin Reg$

contradició

[Idea: analitzar a $\{a^n b^m \mid m \geq 0\}$!!!]

3.1.4 Dado el alfabeto $\Sigma = \{1, 1\}$ demostrar:

$$L_D = \{w \in \Sigma^* \mid |w|_1 = |w|, \wedge \text{ para toda prefixo } u \text{ de } w \mid u|_1 \geq |u|, \}$$

$\uparrow$
Lenguaje de Dyck "0"

es no regular:

Supero L_D es regular. Considero $P = (1^* 1)^*$

Por propiedad intersección L_D de ser regular $L' = P \cap L_D$

$$L' = \{1^n 1^n \mid n \geq 0\}$$

Por pumping

$$L' = \{a^n b^n \mid n \geq 0\} \text{ lo de ser regular, pero}$$

sabemos que no es regular porque $a^n b^n$ No es regular.

$$3.1.1 \quad A = \{a^n b^m \mid m \in \mathbb{N}\}$$

$$L \subseteq A$$

$$L \text{ regular} \iff L \text{ é finit}$$

$\Leftarrow$ L é finit $\Rightarrow$ el podem representar amb n estats, tenim un DFA regular.
Per exemple amb $n=1$

$\Rightarrow$ L é regular $\Rightarrow L$ é finit? Per contradicció

Suposem L é infinit $\wedge L$ é regular, llavors arribem a contradicció.

Si L infinit llavors té una propietat que es repeteix. Llavors pel Lema de Bombeig, si bombejarem a , b o aib trencarem estructura!

3.2 Demuestra que la lenguaje no son reg.

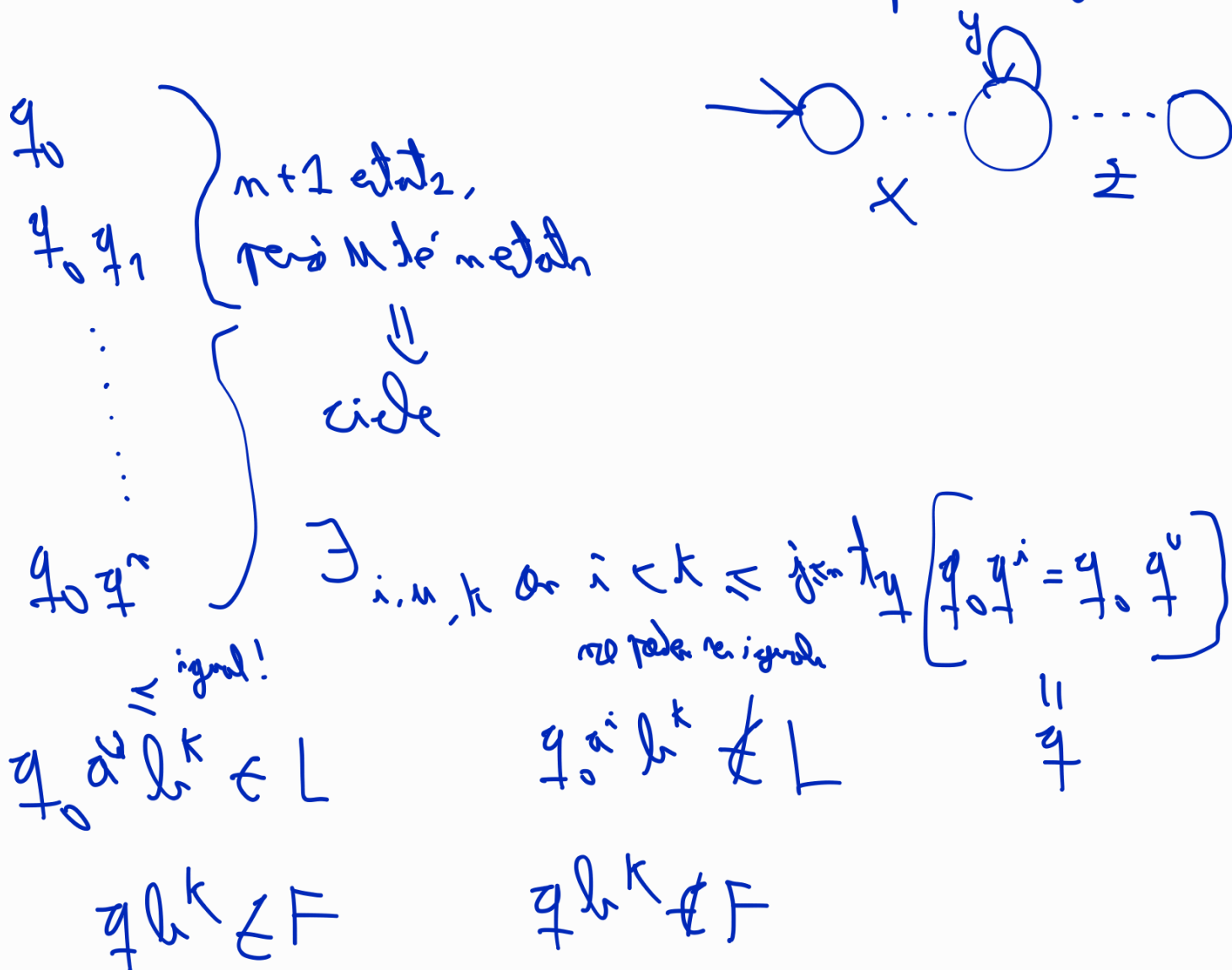
$$b) \{w \in \{a,b\}^* \mid |w|_a = |w|_b\}$$

Suponem L é regular i anillemos a contradicció.

Seu autòmat finit $M = (Q, \Sigma, \delta, q_0, F)$ tal

que $L(M) = L$. Totes le paraules $w \in L$

i $|w| \geq n$ ($n = |Q|$). Havem repetit el estat.



3.2, d)

$$\{w \in \{a, b\}^* \mid |w|_a \neq |w|_b\}$$

Ho da mostrare per induzione. Supponiamo $L_{reg} \Rightarrow$ esiste a_T

$$u = a^n b^{n+n'}$$

$$|w|_a = n, |w|_b = n+n' \Rightarrow |w|_a \neq |w|_b \in L$$

$$|w| \geq n, w = xyz, |xy| \leq n, |y| \geq 1,$$

$$\forall k \geq 0 \quad xy^kz \notin L$$

$$xy^kz = \underbrace{a^n}_{x} \overset{\text{d'almeno}}{\underbrace{a^k}_{y}} \underbrace{a^s}_{z} b^{n+n'}$$

$$\text{per } n+s+m=1$$

$$n+km+s = n+n'$$

$$n-n+km = n+n'$$

$$n+(k-1)n = n+n'$$

$$(k-1)n = 1!$$

$$k-1 = \frac{1}{n}!$$

$$k = \frac{1}{n}! + 1 \geq 2$$

3.2 Demonstra că limba este regulă

$$e) L = \{ w \in \{a, b, c\}^* \mid |w|_a \geq |w|_b \vee |w|_b \geq |w|_c \}$$

Supunem L e regulă. Atunci $w = b^{n+1} c^{n+1} \in L$

Per Pumping $w = xyz$ $|xy| \leq N$

$$x = b^i \quad i \geq 0$$

$$y = b^j \quad |y| \geq 1$$

$$z = b^{N+1-i-j} c^{N+1}$$

(+1 pentru j pe la N :
 $|w|_a \geq |w|_b$!!!)

$$\forall j \geq 0 \quad x y^j z \notin L$$

$$x y^0 z = xz = b^i b^{N+1-i-j}$$

$$e) L = \{ w \in (a,b)^* \mid |w| \in 3 \Rightarrow |w|_a = |w|_b \}$$

System L regular, arithmetic & contrivance!

$$|w|_a \in 3 \wedge |w|_b \in 3$$

Then in DFA A on $|Q| = n$ - more than being

$$\forall w \in L, |w| \geq n$$

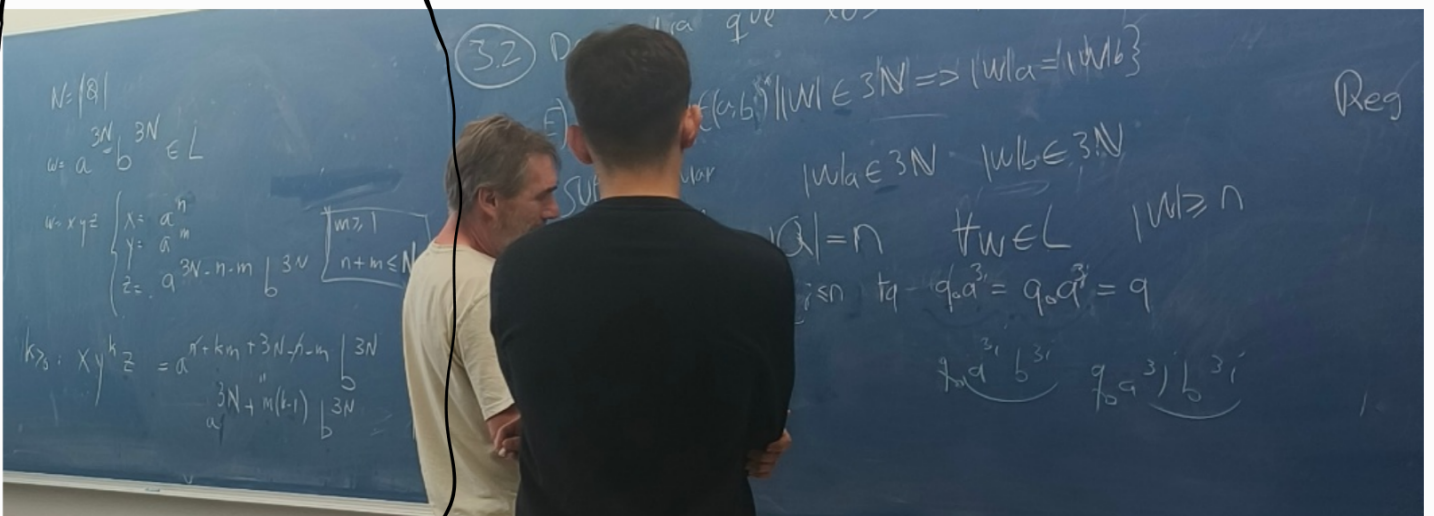
$$\exists i, j \quad 0 \leq \cancel{i} < \cancel{j} \leq n \quad \text{such that } q_0 a^{\cancel{i}} = q_0 a^{\cancel{j}} = q$$

$$q_0 a^{\cancel{i}} b^{\cancel{i}} \in F$$

$$q_0 a^{\cancel{i}} b^{\cancel{j}} \notin F$$

no works!!

no could 3



$$\rightarrow i+2 \in 3$$

$$j+2 \in 3$$

3. $S = \{0, 1, \# \}$ # separator required & universal over $\{0, 1\}$

$$L = \{w \# v \mid w, v \in \{0, 1\}^* \wedge v \text{ mirror } w\}$$

Algne

Handwritten notes on a chalkboard showing a proof for the Pumping Lemma applied to the language $L = \{w \# v \mid w, v \in \{0, 1\}^* \wedge v \text{ mirror } w\}$.

Left side:

- $N = |B|$
- $w = a^{3N} b^{3N} \in L \rightarrow 0 \xrightarrow{x} 0 \xrightarrow{y} 0$
- $w = xyz$ where $x = a^n$, $y = a^m$, $z = a^{3N-n-m} b^{3N}$ with $n+m \leq N$
- $k \geq 0$, $xy^kz = a^{n+km+m+3N-n-m} b^{3N} = a^{k+1+3N} b^{3N}$
- $k=4$

Right side:

- $L = \{u \# v \# z \mid u, v \in \{0, 1\}^* \wedge \text{val}_1(u) + \text{val}_1(v) = \text{val}_1(z)\}$
- $|xy| \leq p$
- $1 \leq i \leq p$
- $2^p - 1 + 0 = 2^p - 1$
- $2^p + 3 > p$
- $p + b = 2^p - 1$
- $a + 2b + c = p$
- contradiction
- $S = XYZ = 1^a 1^b 1^c \# 0 \# 1^p$
- a, b, c
- a, b, c

6.3 a

$\{0101^201^3 \dots 01^n \mid n \in \mathbb{N}\}$ Demonstrate that $\bar{L}$ is not regular

$$h(L) \begin{cases} h(01) = 1 \\ h(11) = 1 \end{cases}$$

$$h(L) = \sum_i 1_{\frac{n(n-1)}{2}} \mid \frac{n(n-1)}{2} \in \mathbb{N}$$

$\uparrow$
 a_i

Then $a_{i+1} > a_i + n$

$$a_{i+1} - a_i = \frac{(n+1)(n+2)}{2} - \frac{n(n+1)}{2}$$

$$a_{i+1} - a_i = \frac{(n+1)(n+2-n)}{2} = n+1$$

Contradiction
Hence $\bar{L}$ is not regular

Q

For R

$$\{1^n 0 1^{n-1} \dots 0 1^2 0 1\}$$

$$1^p 0 1^{p-1}$$

↓

Requires reg₂!!!

$$\{1^k \mid k \text{ prime} \vee k \text{ pow}\} \text{ no Reg}$$

$$a_i = \begin{cases} 2 \cdot i & i \text{ impar} \\ p_i & i \text{ par} \end{cases}$$

12, 3, 1, 5, 6, 7, 8, 11
10, 13, 12, 17

$m \in \mathbb{N}$

$(m+1)^i + 2$
mult 2

$(m+1)^i + 3, (m+1)^i + (m+1)$
mult 3

$$\exists i_m \neq 0 \mid p_{i_m+1} - p_{i_m} \geq m$$

$$\sup |Q| = n \neq 2$$

Let p be no prime $p > n$ (impr)

$$\Rightarrow 1^p \in L$$

$$x = 1^k \quad m \geq 1$$

$$y = 1^m$$

$$z = 1^{p-k-m}$$

$$1^k \quad 1^{i_m} \quad 1^{p-k-m}$$

$$x y^i z = 1^{p+m(i-1)}$$

$$\downarrow$$

$$1^{p+2m} \Rightarrow 1^{p+m(2p+1-1)}$$

$$i-1 = 2p$$

3.8-1) La concatenación de ZL no reg ^(ver 2.8)
 mantiene lo no reg.

$$A = \{w \in \{a, b\}^* \mid |w|_a \geq |w|_b\}$$

$$\Sigma = \{a, b\}$$

$$B = \bar{A} = \{w \in \{a, b\}^* \mid |w|_a \leq |w|_b\}$$

$$A \cdot B = A \cdot \bar{A} = C = \{w \in \{a, b\}^* \mid |w|_a \geq |w|_b \wedge |w|_a \leq |w|_b\}$$

↓

$$C = \{w \mid \exists x, y : w = xy \Rightarrow |x|_a \geq |x|_b \wedge |y|_a \leq |y|_b\}$$

Donc on a:

- Prouver $n^0 a > n^0 b \Rightarrow x \leq w \wedge y \neq \epsilon$
- Prouver $n^0 a = n^0 b \Rightarrow x = w \wedge y = \epsilon$
- Prouver $n^0 a < n^0 b \Rightarrow x = \epsilon \wedge y = w$

$$(L \cup \{\epsilon\}) (\bar{L} \cup \{\epsilon\}) \subseteq \Sigma^*$$

SEMERE!

3.13 Dado lenguaje a par e b par
don un DFA y 2 exp regulares

